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Separation Science and Technology

ISSN: 0149-6395 (Print) 1520-5754 (Online) Journal homepage: http://www.tandfonline.com/loi/lsst20

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To cite this article: SAAIDIA Samia, DELIMI Rachid, BENREDJEM Zahia, MEHELLOU Ahmed, DJEMEL Abdelhak & BARBARI Karima (2017): Use of a PbO_2 electrode of a lead-acid battery for the electrochemical degradation of methylene blue, Separation Science and Technology, DOI: 10.1080/01496395.2017.1291681

To link to this article: http://dx.doi.org/10.1080/01496395.2017.1291681

Accepted author version posted online: 16 Feb 2017.

Use of a PbO_2 electrode of a lead-acid battery for the electrochemical degradation of methylene blue

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Abstract

In this work the electrochemical degradation efficiency of synthetic azo dye, methylene blue, at positive electrode PbO_2 of lead-acid battery was investigated. The structure and morphology of the electrode was investigated by X-ray diffraction, scanning electron microscopy and energy dispersive X-ray spectrometry. The influence of several operating parameter on electro-oxidation of 100 mL of methylene blue solution 100 mg/L was studied. Results indicated that lead-acid battery electrode is effective for removing color and chemical oxygen demand. It is found that current density, the stirring speed and the supporting electrolyte concentration have a positive effect on decolorization and mineralization, and no significant effect of the distance between the electrodes on methylene blue degradation and chemical oxygen demand removal was observed. By contrast, the percentage of color and COD removal decreases with increasing of pH. Kinetic analysis of the results revealed that the chemical oxygen demand removal follows a pseudo first order kinetics.

Key words: lead dioxide electrode, Methylene blue, Electrochemical degradation, Hydroxyl radical, Chemical oxygen demand

1. Introduction

The textile industry, which is a large consumer of water, generates a very important pollution of aqueous medium with discharges highly contaminated with dyes. It has been found that the discharge of dye pollutants into the environment even in a small amount is harmful to the ecological system and human health, so the removal of dye pollutants from wastewaters has received increasing attention over the past decades [1, 2]. The exposure to methylene blue (MB) causes serious injuries to human and animals eyes and in some cases could lead to mental confusion and methemoglobinemia [3]. Various treatment processes have been applied to remove dye from wastewaters, such as biological degradation [4], coagulation–flocculation [5], Fenton's oxidation [6], membrane separation [7] and photocatalysis [8]. However, these processes are quite expensive and have operational problems.

Various innovative technologies have been proposed for the removal of dyes from water. Among these technologies, the electrochemical advanced oxidation processes (EAOPs) constitute the emergent methods for the degradation of organic pollutant. These methods are environmentally friendly and they do not form new toxic wastes [9]. The most popular EAOP for water remediation is anodic oxidation (AO). In AO, the oxidation of organic pollutants is carried out by hydroxyl radicals having high standard reduction potential (2.38 V [10]) produced on the anode surface by oxidation of water. The efficiency of the dyes electrochemical oxidation in aqueous solutions strongly depends on the nature of the electrode used [11]. Various types of electrodes including the active carbon fiber (ACF) [12], Pt [13], Ir [14], Ti/SnO_2 [15], boron-doped diamond (BDD) [16], Ti/RuO_2 [17], IrO_2/TaO_2/RuO_2 [18], RuO_2 [19], PbO_2 [20, 21, 22, 23], Ti/IrO_2/SnO_2/Sb_2O_5 [24] and Ti/SnO_2-Sb [25] were used in dye degradation studies by anodic oxidation.

In AO, the oxidation of organics on the non-active electrodes as SnO_2, PbO_2 and BDD is performed by the physisorbed hydroxyl radical (M ($^{\bullet}$OH)) which is formed as follows:

$$MO_x + H_2O \rightarrow MO_x\,(^{\bullet}OH) + H^+ + e^- \quad (1)$$

However, with the active electrodes like Pt, IrO_2 and RuO_2, oxidation is carried out by the chemisorbed active oxygen as oxygen, included in the network of higher oxide (MO_{x+1}), which is obtained by a reaction between the hydroxyl and the oxygen already present in the oxide as follows [26]:

$$MO_x\,[^{\bullet}OH] \rightarrow MO_{x+1} + H^+ + e^- \quad (2)$$

With, the active electrodes, the organics are converted to carboxylic acids. In contrast, the non-active electrodes lead to mineralization of organics. On the three anodes of high oxygen evolution overvoltage, SnO_2, PbO_2 and BDD, the BDD anode always offers the most advantages regarding removal efficiency and decontamination of wastewaters. However, the application of BDD may be limited due to the high cost of preparation of BDD and fragility of the required electrocatalyst materials. In contrast, the PbO_2 was regarded as an excellent metal oxide electrode and was widely used in electrochemical oxidation due to its low price compared to noble metals, easy preparation, low electrical resistivity, good chemical stability in corrosive media and relatively large surface area [27]. Lead dioxide electrodes, used in the various studies on the electro-oxidation of organic pollutants, were prepared on different substrates, Nb [28], Pb [29], Ti [30], mild steel [31], Ta [32], Ti/SnO_2-Sb_2O_5 [33], C [34] and sometimes they contain dopants such as Bi [35], F [36], Co [37] and Ce [38]. However, no work has so far been published on the electrochemical degradation of organic compounds on positive electrode PbO_2 of lead-acid battery (PE-PbO_2-LAB electrode). This is why in this work we propose to study the feasibility of electrochemical degradation of an azo dye on PE-PbO_2-LAB electrode. Industrial PbO_2 electrode is characterized by X-ray diffraction (XRD) and scanning electron microscopy (SEM). The influence of some operating parameters such as current density, concentration, initial pH, temperature, distance between electrodes and stirring speed on the decolorization and mineralization of MB solution was investigated.

2. Experimental

2.1. Materials

Methylene blue (MB 99% purity) was reagent grade from Merck. Anhydrous sodium sulfate (Na_2SO_4 99% purity) of analytical grade purchased from Sigma Aldrich was used as supporting electrolyte. The solution pH was regulated with sulfuric acid (H_2SO_4 95-97% purity, density 1.83) and sodium hydroxide (NaOH 99% purity), both of analytical grade supplied by Sigma Aldrich. Standard solutions of potassium dichromate ($K_2Cr_2O_7$ 99.9 % purity), Mercury sulfate ($HgSO_4$ 99-100 % purity), sulfuric acid (H_2SO_4) and silver sulfate (Ag_2SO_4 98.5 % purity) were prepared to measure the chemical oxygen demand (COD). These categories were provided by Merck. In this present work, all solutions were prepared with double distilled water. Each analysis is repeated three times and an analysis value corresponds to the average of these three values. The calculated standard deviation is represented in all figures by an error bar.

2.2. Preparation of PE-PbO_2-LAB electrode

The anode used in MB degradation consists of a PE-PbO_2-LAB electrode that was supplied to us by the National Company of electrochemical products (Sétif, Algeria). The principle preparation of positive electrodes is as follows:

Lead balls are oxidized during grinding operation (process called "ball-milling") to air. The resulting dry powder (mixture of PbO and lead free) is attacked to air by H_2SO_4 (1.40 g/cm^3). The formed paste is applied on lead grids which then undergo a maturation under steam at 55°C ("curing") and is then anodically oxidized in a sulfuric acid solution (1.05 g/cm^3) to lead dioxide [39].

2.3. Experimental set-up

The MB electrodegradation experiments were conducted in batch mode in a cylindrical undivided glass reactor containing a 250 mL solution (Figure 1). Anode and cathode consisted of two plates of PE-PbO_2-LAB electrode and stainless steel respectively. The two electrodes (dimensions: 70 mm x 40 mm; thickness: 2 mm) are spaced by 3.5 cm. The temperature of the solution was maintained constant by using a water thermostat (Julabo. Modal EC 5) and the solution were stirred with a magnetic stirrer. All experiments were carried out under galvanostatic conditions using a DC power supply (PS 405 Pro).

2.4. Operating conditions

250 mL of MB solution were used to study the effect of parameters such as the concentration of MB (20-400 mg/L), the current density (5-70 mA/cm^2), the initial pH (3-9) , the concentration of the supporting electrolyte (Na_2SO_4: 2-12.8 g/L), the distance between the electrodes (1-3.5 cm), the stirring speed (200-700 rpm) and temperature (20-50 °C). During the experiments, samples of 5 ml were taken from the reactor at well-defined time intervals, and the concentration of MB and COD were analyzed immediately.

2.5. Instruments and analytical procedures

The solution pH before and after electrolysis was measured with a Extech, EC 500 pH-meter. The color removal was monitored by measuring the decrease in the absorbance at wavelength of 665 nm. Solution mineralization was monitored by measuring chemical oxygen demand COD. COD was determined by the dichromate method using a thermoreactor (WTW CR 2200) and a spectrophotometer. The method consists in oxidizing the organic matter by an excess of potassium dichromate in sulfuric acid medium in the presence of silver sulphate (Ag_2SO_4) and mercuric sulphate ($HgSO_4$) .The solution obtained is heated during 2 h at 150°C. The absorbance value of the excess dichromate is translated into COD using a calibration curve.

The percentage of removal of MB color and COD were calculated using the following Eq. (3) and (4):

$$\%\ Color\ removal = ((A_0 - A_t)/A_0) \times 100 \tag{3}$$

Where A_0 is the absorbance at initial time and A_t that corresponding at time t of the treatment.

And

$$\%\,COD\ removal = ((COD_0 - COD_t)/COD_0\ ) \times 100 \tag{4}$$

Where COD_0 is COD at initial time and COD_t that corresponding at time t of the treatment.

The instantaneous current efficiency (ICE) for the electrochemical oxidation of the dye was calculated considering the values of chemical oxygen demand of the solution at time t and time t+Δt, using the following Eq. (5) [40]:

$$\%ICE = \frac{(COD_t - COD_{t+\Delta t})FV}{8I\Delta t} \times 100 \tag{5}$$

Where COD_t and $COD_{t+\Delta t}$are COD (g/dm^3) at times t and t+Δt respectively, I is the applied current (A), F is the Faraday constant (96487 C/mol), V is the solution volume (L) and 8 is the oxygen equivalent mass (g/equiv).

The energy consumption (EC) for the electrochemical oxidation of MB was calculated in kW.h $(g_{COD})^{-1}$ of treated solution according to Eq. (6) [41]:

$$EC = IUt/(\Delta COD)\,V_s \tag{6}$$

Where I is the applied current (A), U is the cell voltage (V), t is the electrolysis time (h), V_s is the solution volume (L) and ΔCOD is the decay in COD (g/dm^3).

3. Results and Discussion

3.1. Characterization and stability study of electrode

Scanning electron microscope (SEM) was used to analyze the surface morphology of the electrode on a Hitachi S-4800 model instrument. The elemental analysis of the sample was made by EDS using Zeiss type EVO- HD with an EDS detector: Oxford and Aztec software. The phase structure of the PE-PbO_2-LAB electrode was analyzed by X-ray diffraction (XRD) using PAN analytical diffractometer X' pert with Cu K_α = 1.5405980. Cyclic voltammetry measurements were performed at 25 ± 1 °C with a conventional three-electrode cell using a computer-controlled electrochemical workstation (Biologic SP-200). The PE-PbO_2-LAB electrode was used as the working electrode. A platinum wire and saturated calomel electrode (SCE) were used as the counter electrode and reference electrode respectively.

3.1.1. *Chemical analysis*

The concentrations of PbO_2 and $PbSO_4$ in of PE-PbO_2-LAB electrode were respectively determined by volumetric titration of excess oxalate with a $KMnO_4$ solution and by nephelometric analysis of sulfate in the form of $BaSO_4$. Concentrations PbO_2 and $PbSO_4$ in PE-PbO_2-LAB electrode before and after its use as an anode in electrolysis of supporting electrolyte solution (7.2 g /L Na_2SO_4) are shown in the Table 1.

Table 1 shows an increase in the concentration of PbO_2 from 92.39 to 93.88 % and decrease in the concentration of $PbSO_4$ from 2.00 to 1.05 % in electrode after its use in electrolysis of supporting electrolyte solution (7.2 g/L Na_2SO_4) at 30 mA/cm^2.

3.1.2. *Characterization of PE-PbO$_2$-LAB electrode by SEM and EDS*

Figure 2 shows the powder morphology of PE-PbO_2-LAB electrode before (Figure 2a) and after (Figure 2b) its use as an anode in electrolysis of supporting electrolyte solution (7.2 g/L Na_2SO_4). The image (Figure 2a) shows that the electrode of PbO_2 is formed of small grains of PbO_2 assembled between them, forming agglomerates which in turn form aggregates. The powder of the original electrode (Figure 2a) was homogeneous and relatively uniform. In contrast, after electrolysis of supporting electrolyte solution (7.2 g/L Na_2SO_4) for 90 min at 30 mA/cm^2, the presence of pyramid shaped crystal attributed to the β-PbO_2 is clearer (Figure 2b).

EDS is used to identify chemical elements present in the PE-PbO_2-LAB electrode powder and their relative content. The spectra given in Figure 2c, d indicate the presence of C, O and Pb in the electrode powder. We find the same chemical elements in samples before (Figure 2c) and after (Figure 2d) electrolysis. However, the atomic percentages varied (Figures 2c, d). Indeed, the stoichiometry between lead and oxygen changes from 1: 1.85 (before electrolysis) to 1: 1.89 (after electrolysis).

3.1.3. *XRD Characterization*

Figure 3 depicts the XRD patterns of PE-PbO_2-LAB electrode (a) before and (b) after electrolysis in supporting electrolyte solution (7.2 g/L Na_2SO_4). The peaks observed in pattern at 2θ = 25.5°, 32.0°, 36.2°, 49.2°, 52.2°, 59.0°, 62.5°, 67.0° were assigned to the (110), (011), (020), (121), (220), (130), (310) and (022) planes. They are the essential diffraction peaks of tetragonal β-PbO_2. However, the presence of the phase of orthorhombic α-PbO_2 is confirmed by some peaks observed at 2θ =23.3°, 28.6°, 29.9°, 34.3°, 40.6°, 65.1° which were attributed to the (110), (111), (020), (021), (112) and (041) planes. The values of 2θ obtained for α-PbO_2 and β-PbO_2 are in good agreement with the standard data JCPDS. Comparison of the two DRX patterns in Figures 3a, b shows that the intensity of the α-PbO_2 (111) peak decreased after electrolysis (Fig. 3b). The grain size of PbO_2 crystals of electrodes was estimated from the half height width of the strongest diffraction peak using the scherrer equation and was equal to 19 nm.

3.1.4. Characterization of PE-PbO_2-LAB electrode by cyclic voltammetry

It is known that the use of an anode material with high oxygen evolution potential (OEP) is especially desirable since it can decrease the power loss caused by oxygen evolution reaction. The cyclic voltammetric of 0.5 M H_2SO_4 solution at PE-PbO_2-LAB electrode in potential region between 0.2 and 2.5 V at a scan rate 20 mV/s is depicted in Figure 4a. It can be observed that OEP of PE-PbO_2-LAB electrode is 1.51 V (vs. SCE) at 5 mA/cm^2. Hengliang Mo et al [34] have obtained that the OEP values of PbO_2 electrode on different substrates in a 0.03 M Na_2SO_4 solution are roughly close to one another around 1.55 V (vs. SCE).

Figure 4b displays cyclic voltammograms of 7.2 g/L supporting electrolyte in absence and in presence of 20 mg/L MB at 25 °C at a scan rate 20 mV/s. It can be seen that the presence of MB does not bring any difference in voltammetric profile of Na_2SO_4 support electrolyte (Figure 4). Indeed, the polarization curve obtained in the potential range from 0.2 to 2.5 V in presence of MB does not present any new peaks compared with that observed with the supporting electrolyte alone. This could be due to the fact that MB oxidation on PE-PbO_2-LAB surface takes place at a potential very close to that of the electrolyte oxidation. Also it can be seen from Figure 4b that in the presence of MB, current density at the potential 2.4 V is lower compared to that of the supporting electrolyte alone. This result means that the presence of the MB inhibits slightly the kinetic rate of the oxygen evolution reaction.

3.1.5. **Stability of electrode**

3.1.5.1. Stability of the electrode efficiency

To determine the stability of the electrode efficiency, we tested 12 times the PE-PbO_2-LAB electrode in electrolytic experiments of MB solution (100 mg/L MB + 7.2 g/L Na_2SO_4). The results presented in Table 2 show that during the first electrolysis tests, the percentage of color removal increases to stabilize from the fourth electrolysis. This improvement in color removal is likely due to changes in the properties of the PE-PbO_2-LAB electrode. Indeed, comparison of the patterns (Figure 3) before (a) and after (b) electrolysis shows that the peak intensity assigned to β-PbO_2 (110) increased and that which was assigned at α-PbO_2 (111) decreased. The calculation of the ratio of peak areas (S β-PbO_2 / S α-PbO_2) allowed us to confirm this finding. Indeed, the calculated ratio values before (3.52) and after (6.35) electrolysis shows an increase in the ratio value after electrolysis. Given that the β-PbO_2 electrode is more effective than α-PbO_2 electrode in the oxidation of organic molecules [29], we can say that in our case, the conversion of α-PbO_2 in β-PbO_2 in is the one reason of the increase of the electrode efficiency. The comparison of the SEM images (Figures 2a, b) also supports the hypothesis of conversion of α-PbO_2 to β-PbO_2. Indeed, the image (b) corresponding to the electrode after electrolysis shows a stronger presence of grains of pyramid shaped (attributed to the β-PbO_2) than that observed in the image (a). On the other hand, the variation of the concentration of the electrode components can influence the properties of the electrode. Table 1 shows that after electrolysis of supporting electrolyte solution, the percentages of $PbSO_4$ and PbO_2 in the electrode vary. Increasing the percentage of PbO_2 in the electrode (92.39 to 93.88%, Table 1) resulting from the conversion of $PbSO_4$ to PbO_2 is probably the second cause of increase of electrode efficiency.

3.1.5.2. Accelerated electrolysis test

In order to analyze the stability of PE-PbO_2-LAB electrode, the latter has been subjected to current density of 1 A/cm^2 in 1M H_2SO_4 solution. The variation of the cell voltage (U) with electrolysis time is shown in Figure 5. Figure 5 shows that during the initial period, the cell voltage decreases from 3.8 to 3.5 V then stabilizes for a long period. After 600 h the cell voltage begins to increase rapidly and for a short time (3 hours) the electrode is de-activated. We consider that, for a service life equal to 600 h, the PE-PbO_2-LAB electrode is relatively stable. It is important to note that the test period consists of three stages: activation, stabilization and de-activation. The activation is responsible for improving the efficiency of the degradation of MB observed during the first four electrolysis of MB solution (100 mg/L MB + 7.2 g/L Na_2SO_4).

3.1.5.3. Effect of cycles number

The stability of the electrode was also investigated by the submission of the electrode to a large cycle number in a 7.2 g/L Na_2SO_4 solution in the potential range 0.2 to 2.4 V at scan rate of 40 mV/s. From Figure 6 we can see that the current density recorded at 2.4 V for 1, 100th and 230th are 34.7, 34.2 and 33.2 mA /cm^2 respectively. The loss in current intensity after 230 cycles is of 4.3%.

3.2. Effect of experimental parameters on the decolorization and mineralization of MB solution

3.2.1. Effect of MB initial concentration

On account of the fact that the industrial wastewater usually involves different concentrations of MB, it is very significant, from a practical point of view, to investigate the effect of the initial concentration of dye on the performance of the electrochemical oxidation process in the removal of pollutants [42]. This is why we studied the effect of the concentration in a relatively broad range (20-400 mg/L). In Figure 7a can be observed that, for the examined different concentrations, the percentage of color removal increases with time. During the initial 120 min, the percentage of color removal rises rapidly and reaches 95.93%, 90.02%, 69.7%, 55.91% and 48.59% for the initial concentrations of 20, 50, 100, 200 and 400 mg/L, respectively. After 240 min, the percentage of color removal was 99.79%, 99.69%, 98.19%, 89.11% and 78.99%, respectively. We note that the percentage of color removal decreases, with the increase of initial MB concentration. Figure 7b shows that the mineralization of the MB solution is more difficult than its decolorization. In fact, after 240 min of electrolysis, the solutions of 20, 50, 100, 200 and 400 mg/L reach 99.45, 97.21, 90.20, 78.05 and 62.67% of percentage of COD removal respectively. The results can be explained in terms of diffusion control [43], assuming that mineralization occurred on the electrode surface mediated by hydroxyl radicals. At low initial concentrations, the electrochemical reaction is faster than the diffusion. On other hand, more MB is degraded at the interface and more organic acids are expected to be degraded completely. When the initial concentration increases, more organic substances were transferred to the surface of electrode and prevented the contact between pollutant and active sites [40].

3.2.2. Effect of current density

The electrolysis of MB solution was carried out at different current densities (5, 10, 20, 30, 50 and 70 mA/cm^2) at PbO_2 electrode to investigate the influence of current density on the efficiencies of MB degradation and solution mineralization, keeping Na_2SO_4 concentration at 7.2 g/L, MB concentration at 100 mg/L and pH at 3. It can be found that the percentage of color (Fig. 8a) and COD (Figure 8b) increased with increasing the applied current density. The percentage of color removal reach 62.62%, 68.46%, 73.3%, 79.51%, 93.29% and 96.02% at 5, 10, 20, 30, 50 and 70 mA/cm^2 respectively, after 60 min of electrolysis. This demonstrates that the formation rate of the powerful oxidant $^{\bullet}OH$ also increases with current density, which results in the increase of the percentage of color removal. It should be noted that during the initial phase (30 min), the decolorization of the solution is faster at 70 mA/cm^2 and from 60 min of electrolysis; the percentage of color removal of 50 and 70 mA/cm^2 are almost equal. Indeed, comparison of the two Figures 8a, b allows us to say that during the electrolysis process, decolorization and mineralization take place simultaneously. This finding is explained by the non-selective attack of most organic molecules by $^{\bullet}OH$ radicals [44]. We also note that the decolorization process is faster than that of mineralization (Figures 8a, b). This result is explained by the fact that the oxidation of carboxylic acids (obtained in the last step prior to mineralization) is slower than the oxidation of the parent drug and its derivatives. Unlike the percentage of color and COD removal, the ICE decreased with the increase of current density (Figure 8c), indicating that an increase in current density leads to a decrease of the process efficiency. We selected 50 mA/cm^2 for further study, since the difference between the results obtained at 50 and 70 mA/cm^2 is lower, and also because at a density 50 mA/cm^2 decolorization and mineralization are faster with higher percentage of color and COD removal than those obtained with current densities below 50 mA/cm^2.

3.2.3. Effect of initial pH

Solution pH is an important factor for wastewater treatment. The pH values of the solutions were adjusted by adding drops of H_2SO_4 and NaOH. Previous studies have shown that the anodic oxidation of organic compounds is favorable in acidic medium [45]. However, others have shown the opposite; that is to say, the anodic oxidation is favored in alkaline medium [46]. The authors attribute these contradictory results at the difference in the nature of electrolytes, organic compounds and electrodes [47]. In other studies it was found that the effect of pH on the oxidation of organic compounds is negligible [25]. The effect of initial pH values ranging from 3 to 9 on percentage of color and COD removal is illustrated in Figure 9. As seen from the Figure 9a, the percentage of color removal in the acid medium (99.35% at pH 3) is higher than in neutral (94.96% at pH 7) and alkaline medium (90.96 % at pH 9). We also see that the effect of pH on the percentage of COD removal is similar to that for the MB removal, but slightly less pronounced (Figure 9b). The results also showed that the percentage of color and COD removal reached at pH 3, the highest level 99.35%, and 98.68 % respectively. Our results indicate that the elimination of MB and COD is slightly more effective in acidic medium, although the Eq. (7) indicates that the formation of $^{\bullet}OH$ radicals is favored in basic medium. We explain the decrease of the MB removal with the increase of pH by the fact that the increase of pH favors the polymerization reaction of intermediates on the surface of the electrode which that leads to partial passivity of the electrode [48]. In addition, Oxidation potential of hydroxyl radical was higher in acid condition (+2.85 V) than in alkaline condition (+2.02 V) [49].

$$H_2O \longrightarrow {}^{\bullet}OH + H^+ + e^- \quad (7)$$

3.2.4. Effect of supporting electrolyte concentration

In EAOPs sodium sulfate is commonly used as the supporting electrolyte [50]. To study the effect of the concentration of the supporting electrolyte on the percentage of color and COD removal, electrolysis experiments at different concentrations in the range 2-12.8 g/L of Na_2SO_4 were carried out. The results obtained (Figure 10) show that increasing the concentration of the supporting electrolyte of 2 to 7.2 g/L results a slight increase in the percentage of color and COD removal. However, increasing the concentration of the supporting electrolyte of 7.2 to 12.8 g/L leads to a regression the percentage of color removal (99.35 to 98.81%) and COD removal (99.75 to 99.46%). A logical explanation was given for similar results [51]. Indeed, the increase in the concentration of Na_2SO_4 in the range 2-7.2 g/L promotes the formation of peroxodisulfate ions ($S_2O_8^{2-}$) by oxidation of HSO_4^- (Eq. (8)). The peroxodisulfate ions formed oxidize molecules MB and therefore improves the MB removal. However, increasing further the concentration of the supporting electrolyte up to 12.8 g/L, leads to the consumption of the generated hydroxyl radical by high SO_4^{2-} concentration as in Eq. (9) [50].This has led to the regression of the percentage of color and COD removal.

$$2HSO_4^- \longrightarrow S_2O_8^{2-} + 2H^+ + 2e^- \quad (8)$$

$$^{\bullet}OH + SO_4^{2-} \longrightarrow OH^- + SO_4^{\bullet -} \quad (9)$$

3.2.5. Effect of distance between the electrodes

The distance between the electrodes is another parameter that may influence the electrochemical degradation of organic compounds. In order to investigate the effect of this parameter on percentage of color and COD removal, three different distances have been tested (1, 2 and 3.5 cm). The results obtained (Figures 11a, b) show that there is no significant difference between the removals efficiencies obtained with the different distances investigated. However, the variation in the distance between the electrodes has an effect on the voltage to be applied. Indeed, to apply the same value of the current density and keep it constant during the electrolysis experiments for the three distances studied, it was obliged to vary the applied voltage (Table 3). The increase in the voltage is due to the increase in ohmic drop of the electrolyte with the distance between the electrodes. A similar outcome was reported by Fockedey [52]. From Table 3, we find that to impose the same value of current density, when the distance between the electrodes increases it is necessary to increase the applied voltage. Therefore, an increase of the distance between the electrodes leads to increase in consumption of energy (Table 3). Thus, to eliminate MB, applying a distance of 1 cm is as effective as 2 and 3.5 cm, but in addition, it consumes less energy.

3.2.6. Effect of stirring speed

In order to examine the effect of stirring speed of solution and therefore the mass transfer on the electrochemical degradation of MB, the electrolysis experiments were performed at different stirring speed: 200, 400 and 700 rpm. As shown in Figure 12a, percentage of color removal increases with the increase of stirring speed. After a period of 240 min, the percentage of color removal reached 91.84%, 99.35% and 99.49% when the stirring speed was 200, 400 and 700 rpm respectively. From Figure 12a, it is seen that increasing the stirring speed, 200 up to 400 rpm results in a significant increase in the percentage of color removal. However, the increase of stirring speed from 400 to 700 rpm causes only a slight improvement in the percentage of color removal. A similar tendency with percentage of COD removal was observed (Figure 12b). The fact that the increase in the stirring speed improves MB removal, suggests that the mass transfer process plays an important role in the electrodegradation MB at PE-PbO_2 LAB electrode. Similar results were also obtained by Bensalah [53] during the oxidation of synthetic wastewaters containing dyes.

3.2.7. Effect of temperature

The electrolysis of the 100 mg/L MB solution on the PE-PbO_2-LAB anode at an applied current density of 50 mA/cm^2 was examined at four different temperatures in the range 20-50 °C. As seen in Figures 13a, b, the percentage of color and COD removal increase with increasing of solution temperature. The MB degradation at 30, 40 and 50 °C was complete after about 120 min of electrolysis. However, complete mineralization has been observed only for 40 and 50 °C after an electrolysis period of 80 and 120 min respectively. It is estimated that the temperature has little effect on electrochemical oxidation of MB. Effect of temperature on the electrochemical oxidation with $^{\bullet}OH$ radicals has been interpreted in terms of the effect by mediated electro reagents (e.g. peroxodisulphates) [54]. It was mentioned that the oxidation rate of organic compounds with peroxodisulphates ions increases with temperature [55]. Some peroxodisulphates can be formed in solutions containing sulphates during electrolysis at anode according to the following Eq. (10):

$$2SO_4^{-2} \rightarrow S_2O_8^{-2} + 2e^- \quad (10)$$

$S_2O_8^{-2}$ is a powerful oxidizing agent that can oxidize organic matter.

In addition, since the diffusion rate increased with the temperature, an increase in the reaction temperature could bring about an increase in the degradation rate [47]. In addition, the increase in temperature decreases the medium viscosity, and consequently promotes the diffusion rate of the organic material towards the anode surface leading so to an increase in the degradation of organic matter [55].

In fact, as shown in Figures 13, the COD removal follows pseudo first-order kinetics (Eq. (11)) and the apparent rate constant increased with temperature: 0.0122, 0.0218, 0.0301 and 0.0413 min^{-1} at 20, 30, 40 and 50 °C respectively.

$$Ln\,(COD_t/COD_0) = -kt \quad (11)$$

Where COD_0 is the initial COD, COD_t is COD at given time t and k is the kinetic rate constant.

3.2.8. The UV–Vis absorption spectra analysis

Figure 14 shows the evolution of the UV–Visible absorption spectra with time during the galvanostatic electrolysis of the MB solutions. The UV-Visible spectrum recorded in the field 190-800 nm presents four absorption peaks at the wavelengths 246, 290, 612 and 665 nm among which the last corresponds to maximum absorption. The two peaks, 246 and 290 nm are attributed to the aromatic ring and the other two strips 612 and 664 nm can be assigned to the presence of chromophore (dimethylamino group) [56]. Xiaoliang Liang et al ascribed the decrease of the absorption bands at 665 and 602 nm to the N-demethylation of MB [57]. The intensities of these four peaks decrease simultaneously and continuously, and completely disappear after 4 hours of treatment (Figure 14). This implies that the process of demethylation and decomposition of the aromatic rings takes place at the same time. The comparison of (A/A_0) absorbance ratios, calculated at 180 min for the peaks 290 nm (0.3061) and 665 nm (0.1429) shows that the demethylation is faster than the decomposition of the aromatic ring. After 240 min of electrolysis, the peaks at 246 and 290 nm completely disappeared and no new bands appear. This implies that the decomposition by complete oxidation of the methylene blue and its derivatives phenothiazines had occurred.

4. ***Conclusion***

The electrochemical oxidation of MB, using the anodic oxidation process with PbO_2 anode, has been deeply investigated. The characterization by SEM and XRD study of the electrode before and after electrolysis test showed that the PE-PbO_2-LAB electrode consists essentially of tetragonal β-PbO_2 and after electrolysis test the phase β-PbO_2 strengthens it further. The accelerated life test indicated that the service life of PE-PbO_2-LAB electrode reaches 603 h at the current of 1 A. The results of the MB electro-oxidation have shown that the electrode is considered effective in degradation of MB and mineralization of the solution. The main conclusions of electro-oxidation study can be summarized in the following points:

1. The percentage of color and COD removal increases when both dye initial concentration and pH decrease.

2. The increase of current density and the supporting electrolyte concentration in the ranges 5-50 mA/cm^2, 2-7.2 g/L respectively, have a positive effect on the percentage of color and COD removal.

3. The power consumption increases with the distance between the electrodes.

4. Increasing percentage of color and COD removal with the temperature in the studied range is relatively low. Kinetic analysis showed that the COD removal follows pseudo first-order kinetics.

Finally, the anodic oxidation process is an environmentally friendly process to eliminate the toxicity of organic compounds in water. This study demonstrated that the PE-PbO_2-LAB electrode may be used as an effective anode in the electrochemical degradation of organic pollutants.

Acknowledgements

The authors thank the National Company Direction of Electrochemical Products (Sétif, Algeria) for their material support.

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Figures Captions

Figure 1. Experimental set-up

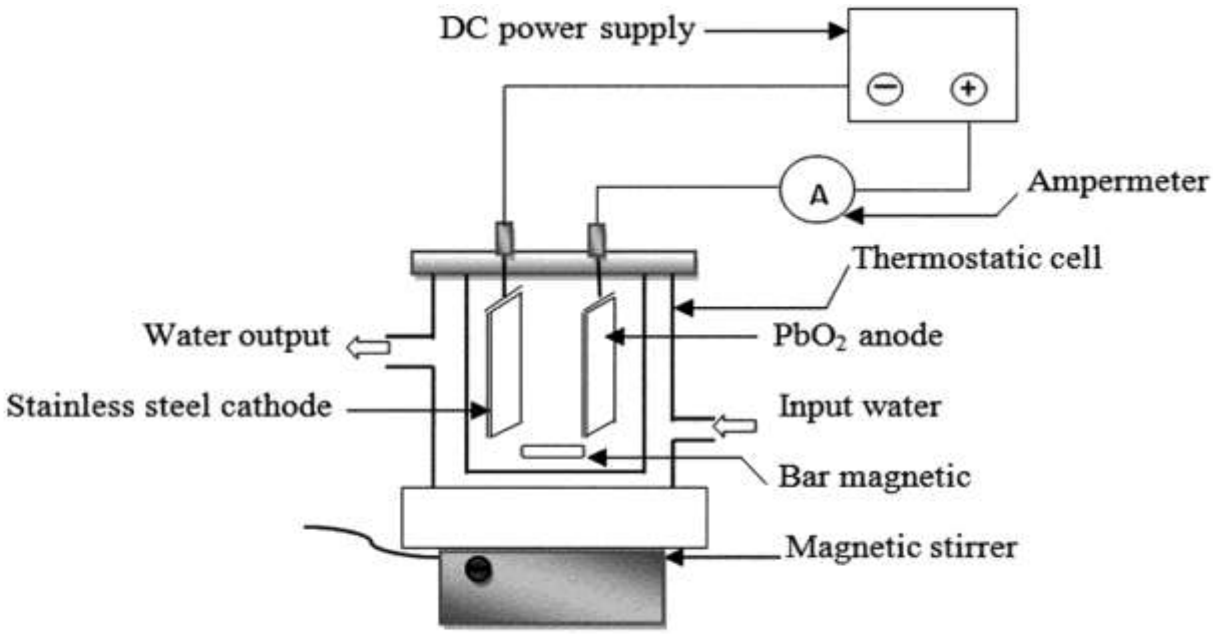


Figure 2. SEM images of PE-PbO_2-LAB electrode **(a)** before electrolysis of supporting electrolyte solution and **(c)** its EDS analysis, SEM images of PE-PbO_2-LAB electrode **(b)** after electrolysis of supporting electrolyte solution and **(d)** its EDS analysis (7.2 g/L Na_2SO_4, current density = 30 mA/cm^2, natural pH and T=30°C)

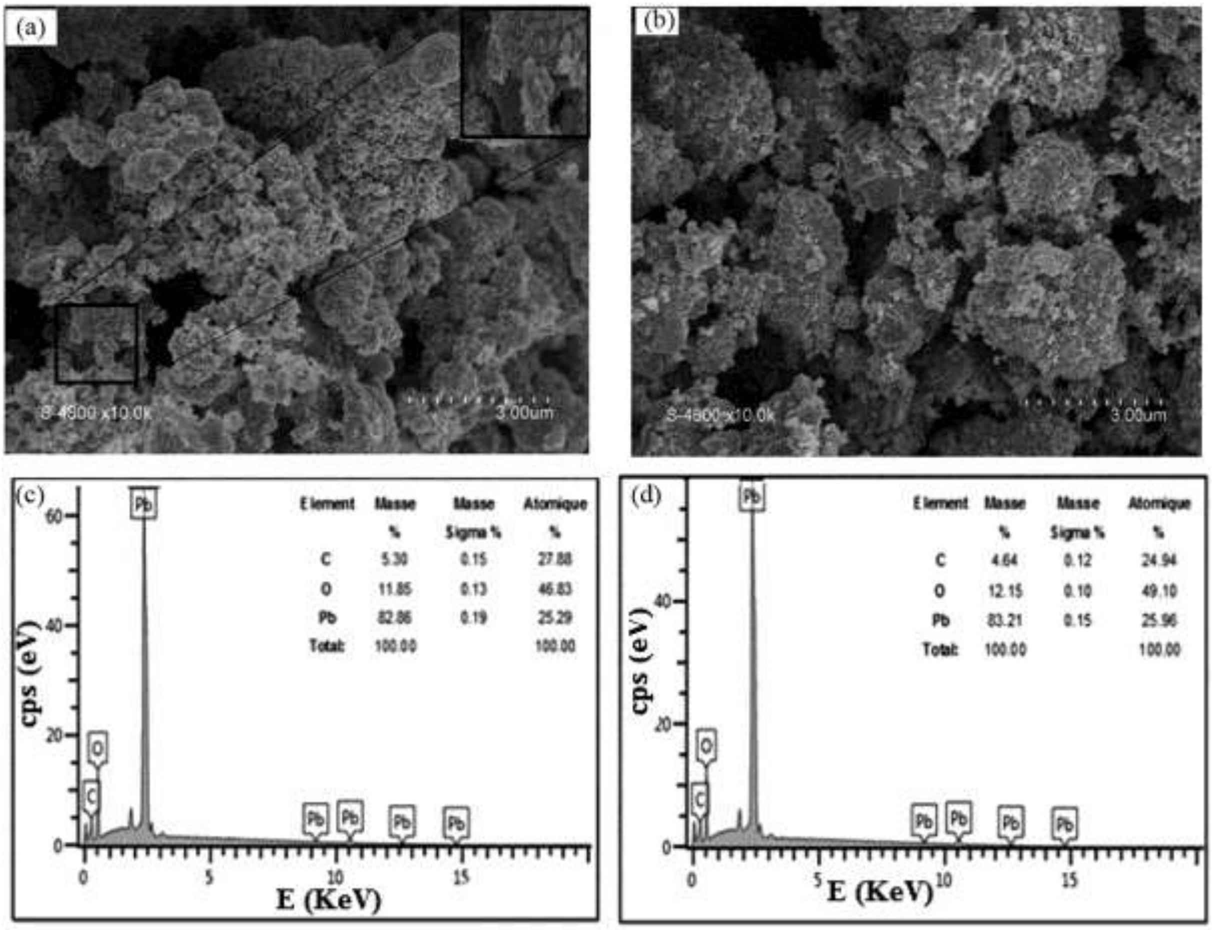


Figure 3. XRD patterns of PE-PbO_2-LAB electrode **(a)** before and **(b)** after electrolysis of supporting electrolyte solution (7.2 g/L Na_2SO_4, current density = 30 mA/cm^2, natural pH and T=30°C)

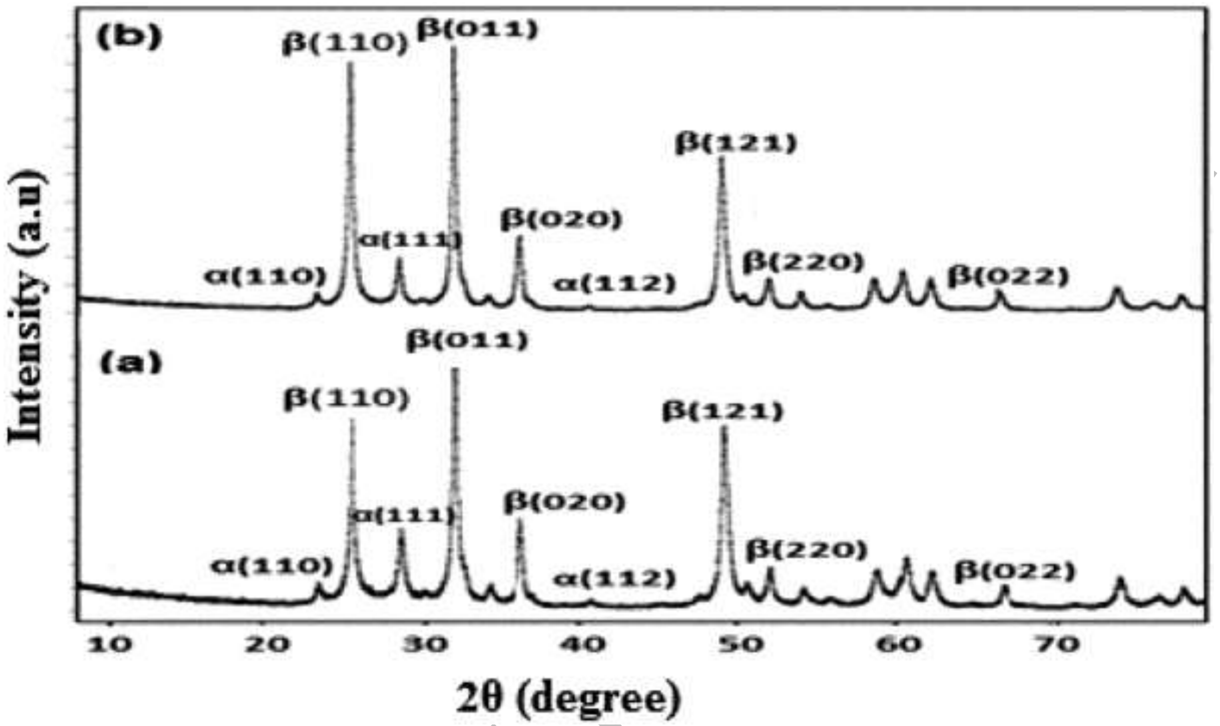
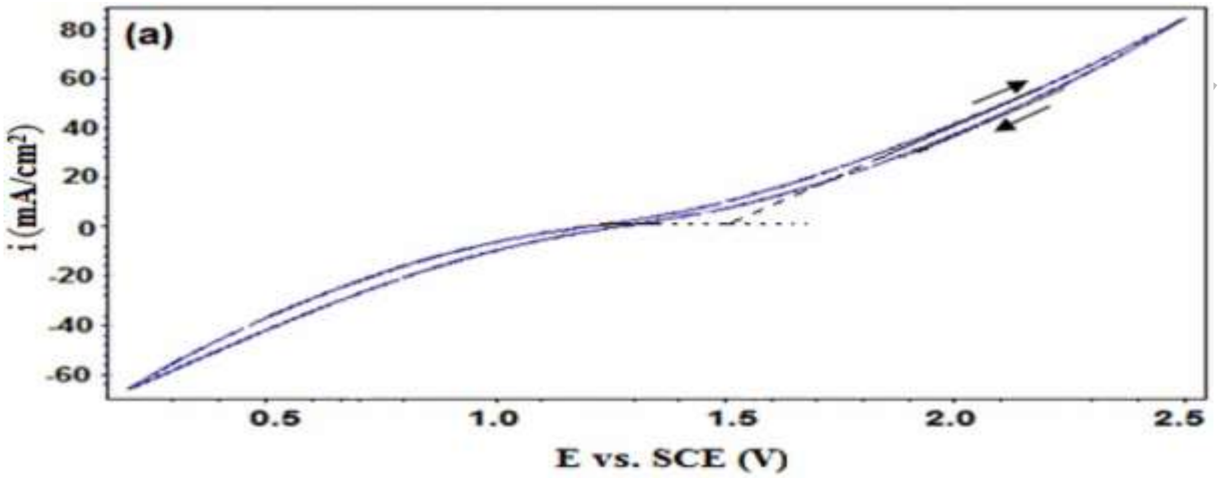


Figure 4. Cyclic voltamogramms of PE-PbO_2-LAB in **(a)** 0.5M H_2SO_4 solution, **(b)** 7.2g/L Na_2SO_4 and 7.2g/L Na_2SO_4 + 20 g/L MB at a scan rate 20 mV/s and T = 20 °C



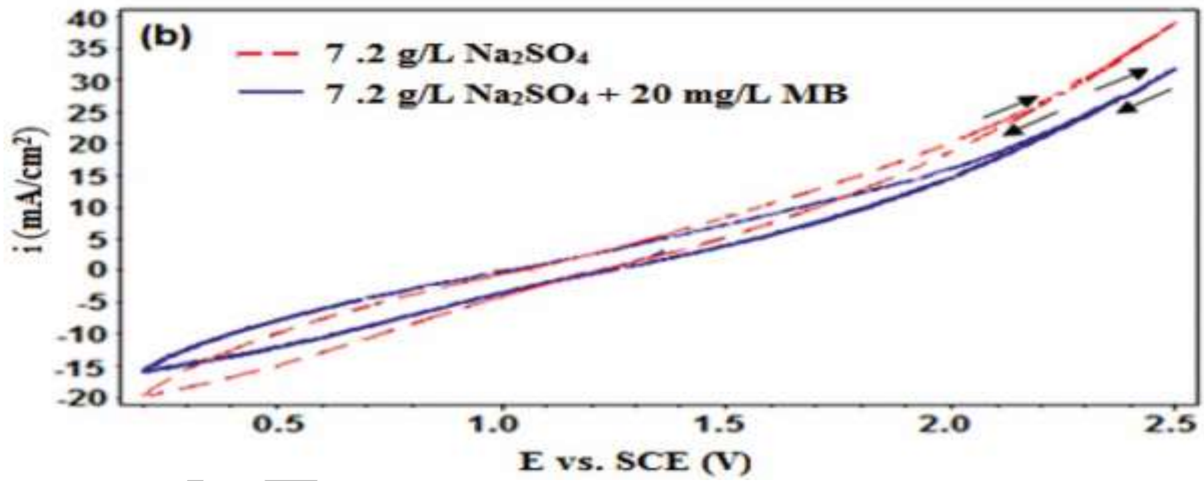


Figure 5. Change of cell voltage with time during the accelerated electrolysis test of PE-PbO_2-LAB electrode in 1M H_2SO_4 solution at 1 A/cm^2 and T = 20 °C

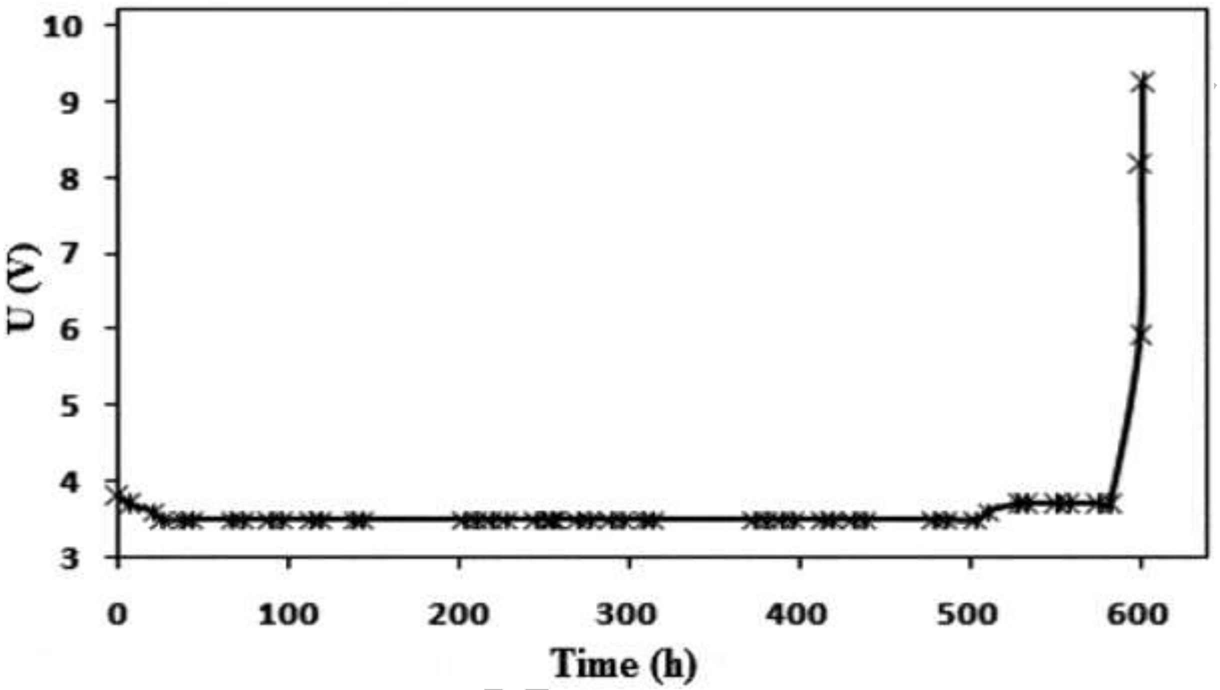


Figure 6. Cyclic voltamogramms of PE-PbO_2-LAB electrode for 230 time consecutive cycles in solution of 7.2 g/L Na_2SO_4 at scan rate of 40 mV/s and T = 20 °C

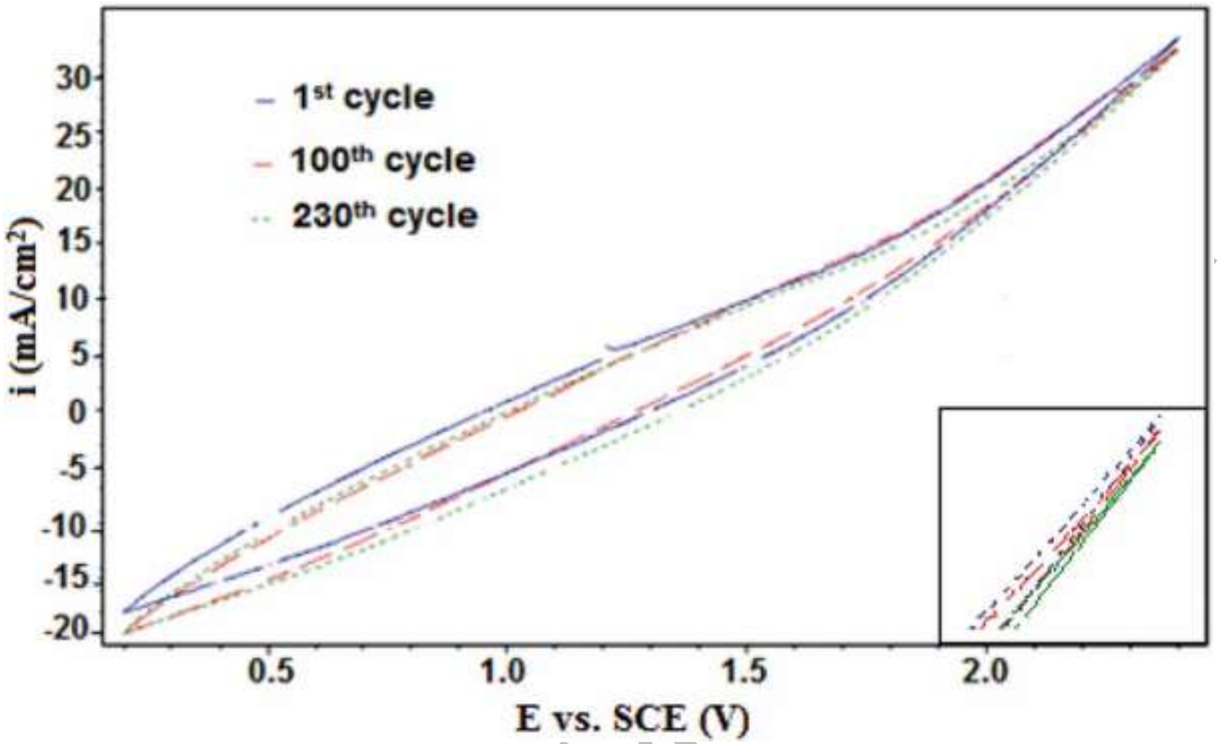


Figure 7. The effect of initial MB concentration on percentage of (a) MB and (b) COD removal with time. (Current density = 50mA/cm^2, natural pH, Na_2SO_4 2 g/L; d=3.5 cm, stirring speed = 400 rpm and T=30°C)

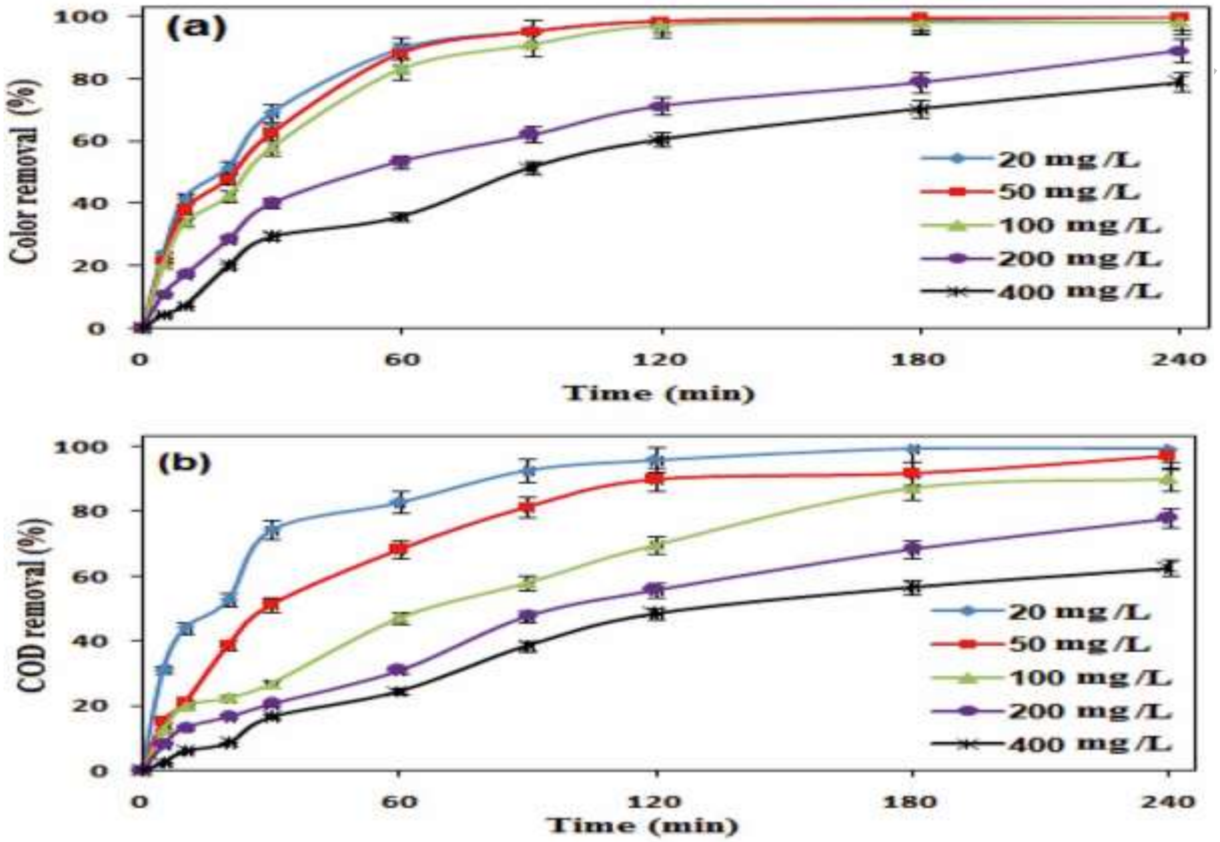


Figure 8. The effect of current density on percentage of **(a)** MB and **(b)** COD removal and **(c)** ICE with time. (MB concentration = 100 mg/L, natural pH, Na_2SO_4 2 g/L, d =3.5cm, stirring speed = 400 rpm and T=30°C)

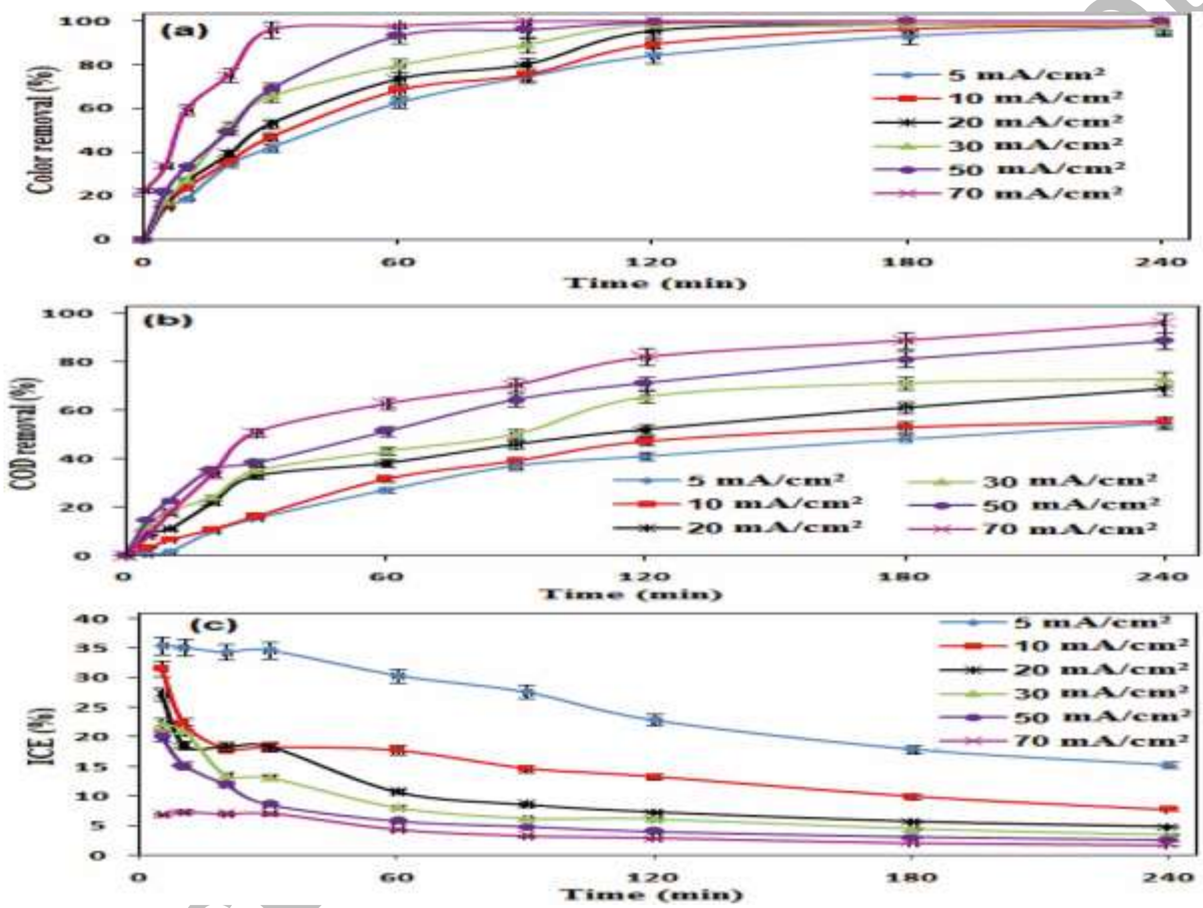


Figure 9. The effect of pH on percentage of **(a)** MB and **(b)** COD removal with time. (MB concentration = 100 mg/L, current density = 50 mA/cm^2, Na_2SO_4 2 g/L, d = 3.5cm, stirring speed = 400 rpm and T=30°C)

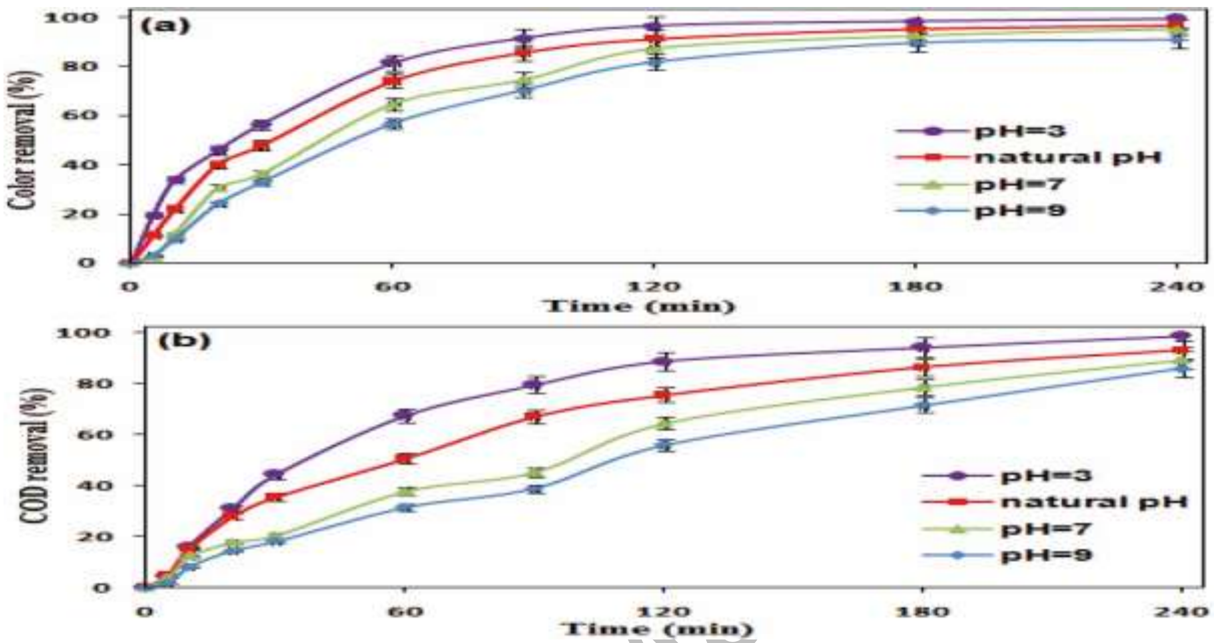


Figure 10. The effect of concentration supporting electrolyte (Na_2SO_4) on percentage of **(a)** MB and **(b)** COD removal with time.(MB concentration = 100 mg/L, current density = 50 mA/cm^2, pH=3, d =3.5cm, stirring speed = 400 rpm and T=30°C)

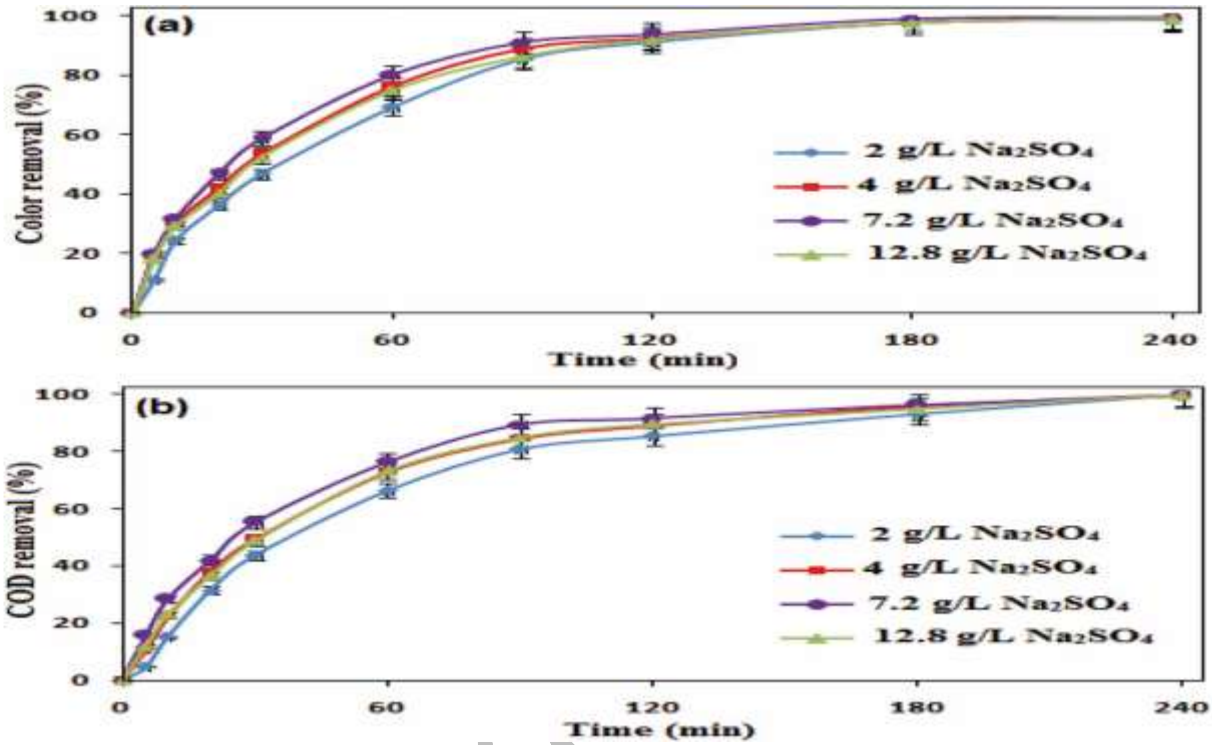


Figure 11. The effect of distance between the electrodes on percentage of **(a)** MB and **(b)** COD removal with time. (MB concentration = 100 mg/L, current density = 50 mA/cm^2, pH=3, Na_2SO_4 7.2g/L, stirring speed = 400 rpm and T=30°C)

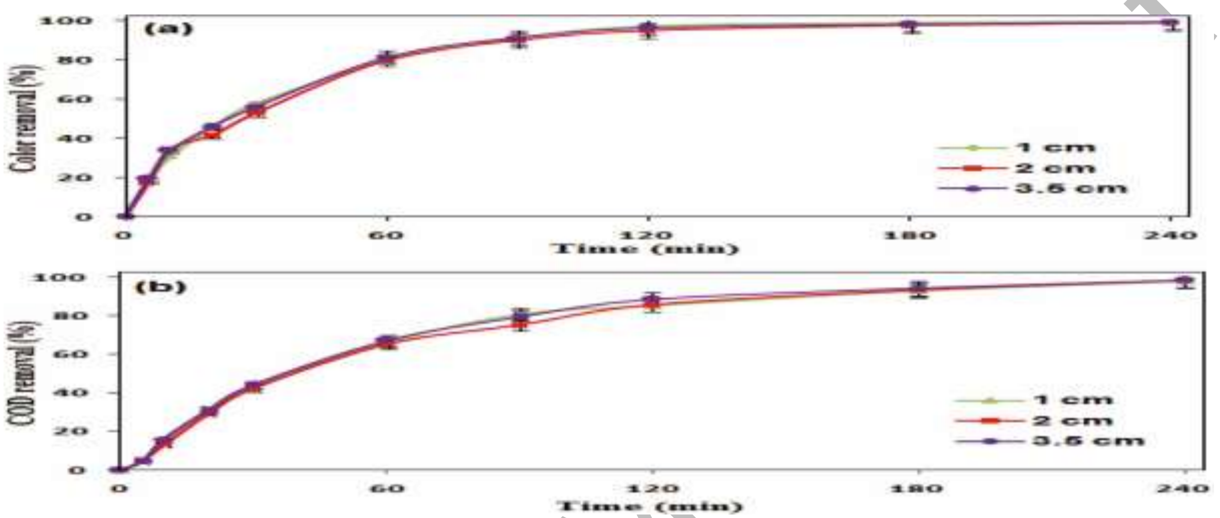


Figure 12. The effect of stirring speed on percentage of **(a)** MB and (**b)** COD removal with time. (MB concentration = 100 mg/L, current density = 50 mA/cm^2, pH = 3, Na_2SO_4 = 7.2 g/L, d =1cm and T = 30°C)

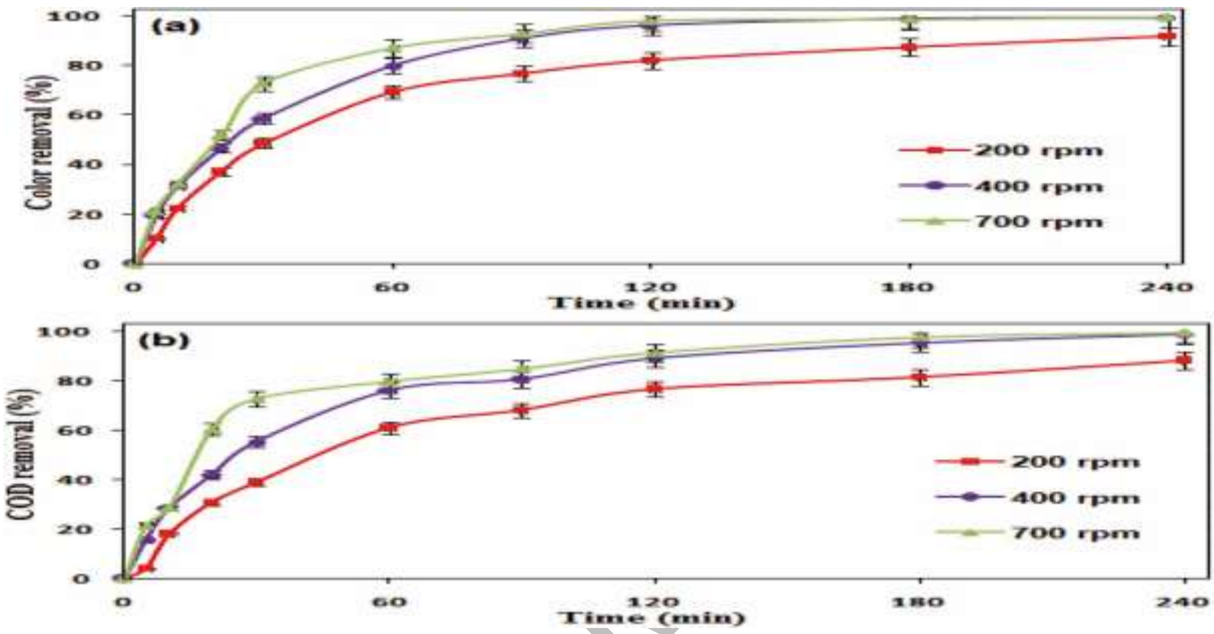


Figure 13. The effect of the temperature on percentage of **(a)** MB and **(b)** COD removal with time, **(c)** Linear regression for COD removal with time during the electrochemical oxidation of MB on the PE-PbO_2-LAB anode for different temperature (MB concentration = 100 mg/L, current density = 50 mA/cm^2, pH = 3, Na_2SO_4 7.2 g/L, d = 1 cm)

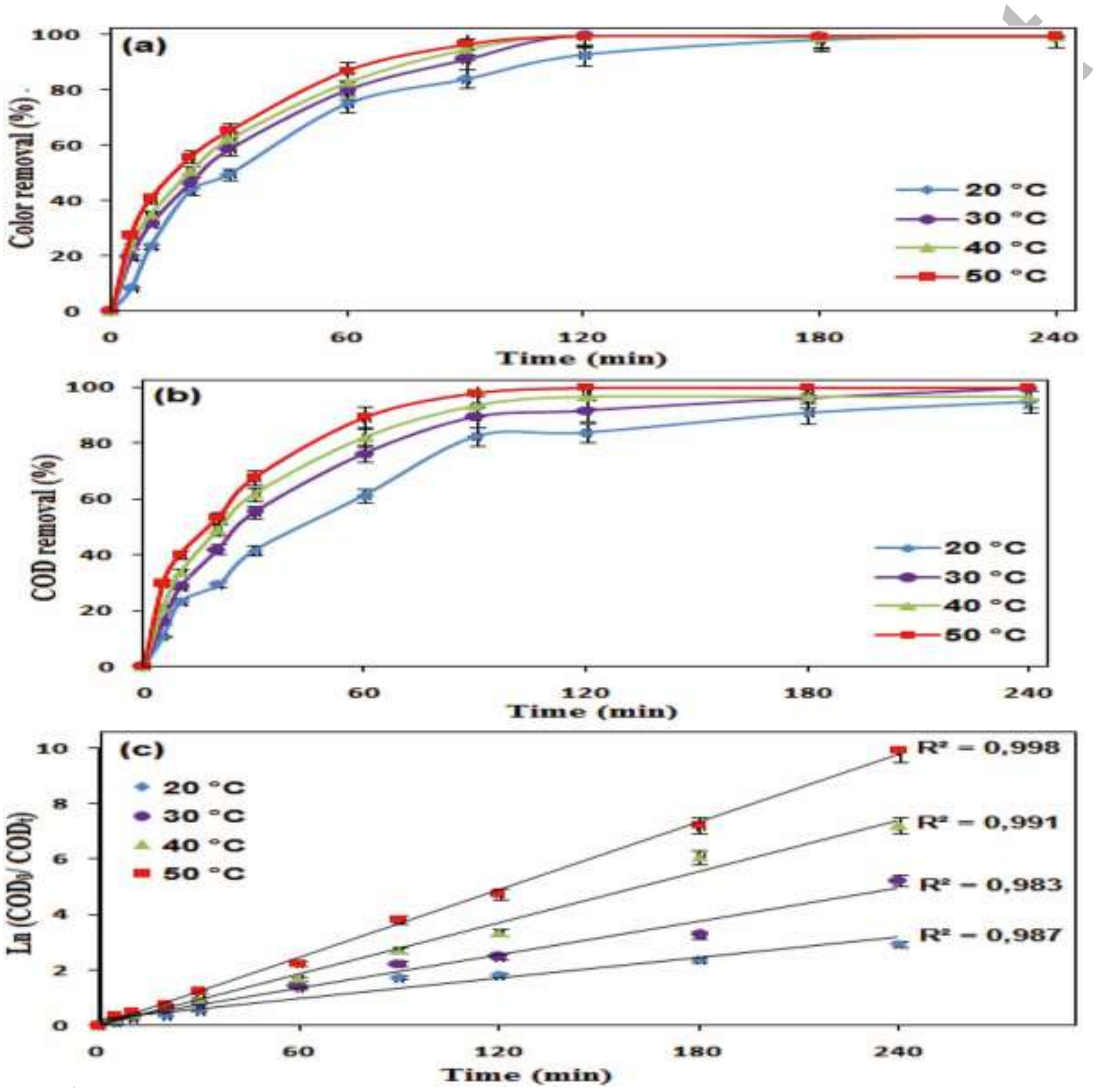


Figure 14. Evolution of the MB absorption spectrum with the reaction time. (MB concentration = 100 mg/L, diluted 25 times, current density = 50 mA/cm^2, pH = 3, Na_2SO_4 7.2 g/L, d =1cm and T = 30°C)

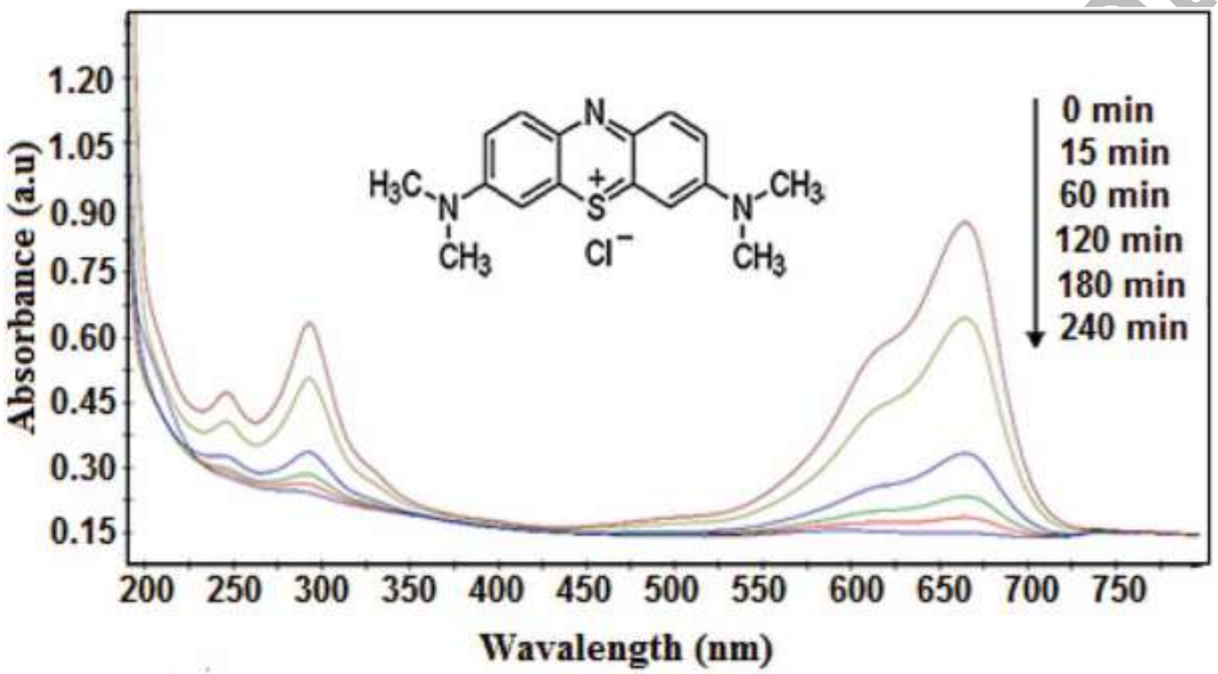


Table 1. Chemical analysis results of PE-PbO_2-LAB electrode before and after electrolysis of supporting electrolyte solution (7.2 g/L Na_2SO_4, current density = 30 mA/cm^2, natural pH and T=30°C).

	PbO_2 (%)	$PbSO_4$ (%)
Before electrolysis	92.39	2.00
After electrolysis	93.88	1.05

Table 2. The stability of PE-PbO_2-LAB electrode in electrolytic experiments of MB solution (MB concentration = 100 mg/L, Na_2SO_4=7.2 g/L, current density = 30 mA/cm^2, natural pH, d =1 cm and T=30°C).

Experiment number	1	2	3	4 -12
Color removal efficiency (%)	81.24	89.35	94.60	95.39 – 95.55

Table 3. Effect of distance between electrodes

Distance (cm)	A (mA/cm^2)	U (V)	EC ($kwhg_{COD}^{-1}$)
1	50	18.93	2.42
2	50	25.33	3.03
3.5	50	30.5	3.64